

Intro



- In many games, there are built in “trainers” to help you learn/practice
- Games have set difficulty, such as easy, medium, hard
- These set difficulties may be too difficult/easy for the player
- What if there was a way to have an adaptive trainer/player that would be a perfect match in terms of difficulty
- Using Connect 4, harder than tic-tac-toe, easier than chess

Results

- With RAG, the responses were very knowledgeable, but usually printed knowledge that wasn't familiar to a new user
- The response times were quite long from Gemini, but overall everything worked

When asking : What's the best move for me right now?

"Have you noticed any patterns forming in the center of the board?"

***RAG - Retrieves documents to generate better answers**

Objective

- Create an adaptive player/trainer in connect 4 using the Gemini LLM

Setup



- Develop a simple Connect 4 game in Python
- Add an intelligent opponent using AI
- Using the Gemini LLM (from Google), implement it into code using the API key
- Use Gemini as a tutor, giving help when prompted
- Instead of giving the actual answer, Gemini will ask user for what the best play is, and gives its thoughts based on that

Conclusions



- Combines Connect 4 game with AI features
- Includes an AI that gives tough competition
- Has an AI Tutor that helps players learn using smart questions (Socratic method) and research (RAG)
- Compared to a human tutor, the system uses its extensive knowledge base through Gemini to analyze game states and provide strategic feedback almost instantly.

References

Prof. Robert Chun, Google Gemini, SJSU King Library